Situation & Task (The Problem)

As Supervisor, Onsite IT & Automation Systems, I identified a gap in how we handled password and account expiration for ~2,600 users in on-premises Active Directory, synced to Entra ID through Entra Connect. Expirations were handled reactively: people usually found out when they could no longer sign in. Some of this was done by hand, but only on a very limited scale. Covering all ~2,600 users by hand would have taken an **estimated 380 hours a year**, based on industry-standard effort models for identity lifecycle operations, so it **could not be sustained at that scale**. That gap left the organization exposed to an estimated **up to 1,800 avoidable expiration-related service desk tickets**.

Action (The Solution)

I scoped and built **scheduled, unattended PowerShell automation** against on-premises Active Directory that warns users, and their managers where needed, before a password or account expires.

- **Automated Detection:** Scheduled runs query AD for passwords and accounts approaching expiration within defined warning windows.
- **Proactive Communication:** Built a manager-aware notification framework that sent **9,000+ proactive notifications**, so users, and managers where their action was needed, could act before access was lost.
- **Policy Alignment:** Warning windows and rules follow the organization's Security policies for password and account expiration.
- **Audit-Ready Logging:** Every run and every notification is logged, so it is always possible to show who was warned and when.

Result (The Impact)

The "Zero-Touch" initiative replaced reactive lockouts with proactive warnings, with no ongoing manual effort from IT:

- **Effort Covered:** An estimated **380 hours a year** of manual tracking and follow-up handled by automation.
- **Risk Reduction:** An estimated **up to 1,800 service desk tickets** avoided, with a full audit trail of every warning sent.